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United States Patent	12384603
Kind Code	B2
Date of Patent	August 12, 2025
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Sealing cap for a container closure and container closure comprising such a sealing cap

Abstract

A sealing cap made of an elastic plastic material for a container closure includes a container socket base formed on a container for disposing a container socket having a bung plug disposed in a bung socket formed on the container socket base, the sealing cap comprising a lid disk which has a fastening means disposed on an underside of the lid disk and provided with a snap-lock means for forming a snap-lock connection with the bung plug, and an edge skirt disposed at the circumference of the lid disk and connected to the lid disk via a predetermined breaking means. The lid disk has a support means for supporting the lid disk on the container socket base, the support means being disposed radially inside adjacent to the predetermined breaking means on the underside of the lid disk.

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Appl. No.:	18/025822
Filed (or PCT Filed):	July 08, 2021
PCT No.:	PCT/EP2021/069054
PCT Pub. No.:	WO2022/053202
PCT Pub. Date:	March 17, 2022

Prior Publication Data

Document Identifier	Publication Date
US 20230286705 A1	Sep. 14, 2023

Foreign Application Priority Data

DE

10 2020 123 861.4

Sep. 14, 2020

Publication Classification

Int. Cl.: **B65D41/58** (20060101); **B65D39/02** (20060101); **B65D39/08** (20060101); **B65D55/08** (20060101)

U.S. Cl.:

CPC **B65D41/58** (20130101); **B65D39/025** (20130101); **B65D39/082** (20130101); **B65D55/0863** (20130101); B65D2401/25 (20200501)

Field of Classification Search

CPC: B65D (41/04); B65D (41/58); B65D (39/025); B65D (39/082); B65D (51/18); B65D (55/0863); B65D (2251/0075); B65D (2251/0015); B65D (2401/25); B65D (2101/0023)

USPC: 220/789; 220/661; 215/255

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
1847108	12/1931	Hirsch et al.	N/A	N/A
4320861	12/1981	Rieke	222/541.9	B65D 55/0863
4883194	12/1988	Fernandes	220/296	B65D 39/08
5156511	12/1991	De Smet	N/A	N/A
5232120	12/1992	Dunken et al.	N/A	N/A
5649639	12/1996	Dolviet	220/203.11	B65D 51/1644
6112922	12/1999	Nofer et al.	N/A	N/A
6360908	12/2001	Kline	220/276	B65D 51/18
6968968	12/2004	Buckley	220/601	B65D 55/08
7690525	12/2009	Schutz	220/259.3	B65D 47/121
7958995	12/2010	Dedmon et al.	N/A	N/A
8727175	12/2013	Schutz	220/601	B65D 39/082
10683147	12/2019	Choltco-Devlin	N/A	B65D 47/265
2004/0045965	12/2003	Dwinell	220/257.2	B65D 39/08
2005/0082076	12/2004	Przytulla	174/50	B65D 1/12
2005/0092751	12/2004	Alvares	220/276	B65D 21/0219
2005/0121451	12/2004	Van Bruggen	220/257.2	B65D 51/18
2006/0096985	12/2005	Stolzman	220/259.2	B65D 51/1644
2010/0000998	12/2009	Schutz	220/288	B65D 39/082
2010/0282705	12/2009	Ledemeney	215/251	B65D 41/46

2010/0282751	12/2009	Quirijnen	220/303	F16K 24/06
2010/0301046	12/2009	Schuetz	N/A	N/A
2016/0272385	12/2015	Klatt	N/A	N/A
2016/0297582	12/2015	Julien	N/A	B65D 41/083
2018/0056257	12/2017	Busch et al.	N/A	N/A
2018/0290800	12/2017	Filho et al.	N/A	N/A
2018/0319548	12/2017	Pree	N/A	N/A
2019/0344942	12/2018	Klatt	N/A	N/A
2020/0290789	12/2019	Weyrauch	N/A	N/A
2020/0391908	12/2019	Klatt	N/A	B65D 39/086

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
200500626	12/2005	CL	N/A
3532925	12/1985	DE	N/A
9102424	12/1990	DE	N/A
4041536	12/1991	DE	N/A
102017223074	12/2018	DE	N/A
2607136	12/2012	EP	N/A
2008139196	12/2007	WO	N/A
2019218023	12/2018	WO	N/A

OTHER PUBLICATIONS

PCT International Search Report and Written Opinion, PCT/EP2021/069054, Oct. 19, 2021, 15 pages. cited by applicant

China National Intellectual Property Administration, Notice of the First Review Opinion and Search Report, Application No. 202180062585.8, Sep. 14, 2024, 7 pages [English Language Translation Only]. cited by applicant

Republic of Colombia, Superintendency of Industry and Commerce, Office Action and Search Report, Application No. NC2023/0004431, Sep. 12, 2024, 7 pages [English Language Translation Only]. cited by applicant

Canadian Intellectual Property Office, Office Action, U.S. Pat. No. 3,191,869, Aug. 13, 2024, 3 pages. cited by applicant

Japan Patent Office, Notice of Reasons for Refusal, Application No. 2023-515780, Mar. 5, 2024, 22 pages. cited by applicant

National Institute of Industrial Property—INAPI—Chile, Expert Report on Invention Patent Application, Application No. 202300643, Jun. 14, 2024, 10 pages [English Language Translation Only]. cited by applicant

Australian Government—IP Australia, Examination Report No. 2, Application No. 2021338896, Nov. 13, 2024, 3 pages. cited by applicant

Australian Government—IP Australia, Examination Report No. 3, Application No. 2021338896, Nov. 28, 2024, 3 pages. cited by applicant

Egyptian Patent Office, Office Action, Application No. 2023020246, Dec. 1, 2024, 5 pages [English Language Translation Unavailable]. cited by applicant

Korean Intellectual Property Office, Office Action, Application No. 10-2023-7011857, Feb. 12, 2025, 12 pages. cited by applicant

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Background/Summary

- (1) The present application is a national stage application filed under 35 U.S.C. § 371 of International Application No. PCT/EP2021/069054, filed Jul. 8, 2021, which claims priority to German Patent Application No. 10 2020 123 861.4, filed Sep. 14, 2020. Each of the aforementioned patent applications is incorporated by reference in its entirety for all purposes.
- (2) The present disclosure relates to a sealing cap made of an elastic plastic material for a container closure comprising a container socket base formed on a container for disposing a container socket having a bung plug disposed in a bung socket formed on the container socket base, the sealing cap comprising a lid disk which has a fastening means disposed on the underside of the lid disk and provided with a snap-lock means for forming a snap-lock connection with the bung plug, and an edge skirt disposed at the circumference of the lid disk and connected to the lid disk via a predetermined breaking means. Moreover, the disclosure relates to a container closure provided with such a sealing cap.
- (3) A sealing cap of the kind described above is also known from WO 2008/139196 A1. The known sealing cap, which is made of a dimensionally elastic plastic, has a snap-lock means which is disposed on the underside of the lid disk and by means of which the sealing cap is connected to a bung plug disposed in a bung socket of a container closure. The lid disk, on whose outer circumference an edge skirt connected to the lid disk via a predetermined breaking means is disposed, protrudes beyond the opening edge of the bung socket by a lid edge to which the edge skirt is connected via the predetermined breaking means. A destruction of the predetermined breaking means including an at least partial detachment of the edge skirt from the lid disk indicates that there has been an attempt to remove the sealing cap from the container closure.
- (4) The lid disk and the lid edge of the lid disk of the known sealing cap are spaced apart from the bung socket. As a result, the predetermined breaking means are destroyed not only if there is an attempt to remove the sealing cap from the container closure by reaching below the edge, but also if pressure is exerted on the lid disk from above that causes a deformation of the lid disk which is transmitted to the edge skirt via the predetermined breaking means with the result that the edge skirt rests on the wall surface of the container surrounding the bung socket. This type of load can occur in particular if containers provided with container closures are stacked improperly and the upper container placed on the lower container exerts pressure on the lid disk of the sealing cap. Damage caused to the predetermined breaking means in this way has the consequence that the manipulation protection intended by the sealing cap no longer exists and a potential subsequent manipulation of the container closure will therefore no longer be detectable.
- (5) Hence, the object of the present disclosure is to propose a sealing cap of the kind mentioned above that has improved functional reliability.
- (6) To attain this object, the sealing cap according to the disclosure has the features of claim 1.
- (7) According to the disclosure, the lid disk of the sealing cap has a support means for supporting the lid disk on the container socket base, the support means being disposed radially inside adjacent to the predetermined breaking means on the underside of the lid disk.
- (8) The support means disposed on the lid disk radially inside adjacent to the predetermined breaking means of the sealing cap according to the disclosure has the effect that once the sealing cap has been disposed on the container closure and the snap-lock connection between the bung plug has been established, there is a force fit between the support means and the container socket base of the container socket with the result that a deformation of the lid disk in the direction of the socket base due to external pressure has the effect that the outer load is discharged into the

container socket base via the support means and is not introduced into the edge skirt via the predetermined breaking means.

(9) In contrast, the fact that the lid edge of the lid disk of the known sealing cap is spaced apart from the bung socket causes the lid edge together with the predetermined breaking means disposed on the lid edge and the edge skirt connected thereto to be deflected in the direction of the container socket base, which means that the predetermined breaking means may be damaged when the edge skirt comes into contact with or rests on the container socket base. The edge skirt can be in contact with or rest on the container socket base directly or via deposits formed on the container socket base and allowing forces to flow between the edge skirt and the container socket base, deposits of this kind easily occurring during storage and transport of containers of this kind. This is particularly true if the containers are IBCs, which are typically transported and stored repeatedly.

(10) Preferably, the support means is an edge web disposed concentrically with the edge skirt on the underside of the lid disk so that the supporting effect can be effective continuously across the entire circumference of the sealing cap.

(11) If the edge web has a lower edge recessed relative to a lower edge of the edge skirt with respect to the underside of the lid disk, access to the lower edge of the edge web for removing the sealing cap from the bung socket is prevented since the edge skirt is disposed in such a manner that it covers the lower edge of the edge web.

(12) If the snap-lock means of the fastening means is connected to a pull tab disposed at the circumference of the lid disk via a connection means, the connecting means having a lid strip which is formed in the lid disk and separated from adjacent areas of the edge skirt, of the edge web and of an inner part of the lid disk by lateral predetermined breaking means and connected to the snap-lock means via a pull web, the sealing cap can preferably be completely removed from the bung plug without the edge web acting as the support means preventing the removal after the predetermined breaking means formed on the edge skirt has been damaged.

(13) In a first alternative, the container closure according to the disclosure includes a container socket base that is a lid for a container opening formed in a container wall, the lid comprising the bung socket which is disposed in a lid disk of the lid and which penetrates the lid disk in such a manner that the support means formed on the underside of the lid disk of the sealing cap rests on a top of the lid disk of the lid and the edge skirt is spaced apart from the top of the lid disk of the lid.

(14) According to the disclosure, the container socket base is a lid for a container opening formed in a container wall, the lid comprising the bung socket which is disposed in a lid disk of the lid and which penetrates the lid disk in such a manner that the support means formed on the underside of the lid disk of the sealing cap rests on a top of the lid disk of the lid and the edge skirt is spaced apart from the top of the lid disk of the lid.

(15) Preferably, the bung socket has an opening edge protruding from the top of the lid in such a manner that the support means rests on the opening edge of the bung socket.

(16) A container closure according to the disclosure configured in this way allows the sealing cap according to the disclosure to be used on an IBC.

(17) According to another alternative, the container closure according to the disclosure has a container socket base that is a wall portion of a container, the wall portion comprising a bung socket formed in the wall portion in such a manner that the support means formed on the underside of the lid disk of the sealing cap rests on a wall surface of the container and the edge skirt is spaced apart from the wall surface of the container.

(18) The container socket base of the container closure configured according to the disclosure is a wall portion of a container, the wall portion comprising a bung socket formed in the wall portion in such a manner that the support means formed on the underside of the lid disk of the sealing cap rests on a wall surface of the container and the edge skirt is spaced apart from the wall surface of the container.

(19) A container closure configured according to the disclosure in this way allows the sealing cap according to the disclosure to be used on any container, in particular a drum.

Description

- (1) A preferred embodiment of the disclosure will be discussed in more detail below with reference to the drawing.
- (2) FIG. 1 is an isometric illustration of an IBC comprising a container closure disposed on the top of an inner container of the IBC;
- (3) FIG. 2 is an individual illustration of the container closure illustrated in FIG. 1;
- (4) FIG. 3 is a section view of the container closure illustrated in FIG. 2;
- (5) FIG. 4 is a section view of the container closure disposed on a container opening of the IBC illustrated in FIG. 1 along line IV-IV in FIG. 1;
- (6) FIG. 5 is a bottom view of the sealing cap of the container closure illustrated in FIG. 2.
- (7) FIG. 1 shows a transport and storage container **10**, which is also referred to as an IBC and which has an inner container **13** disposed on a transport pallet **11** and accommodated in a cage **12**, inner container **13** being provided with a container fitting **14** on a front side and with a container opening **16**, which can be closed with a container closure **15**, at a top.
- (8) As illustrated in FIGS. 2 to 4, container closure **15** has a lid **17**, which is a screw top in the case at hand and screwed to an opening edge of container opening **16**. A bung socket **19** is disposed in a lid disk **18** of lid **17**, bung socket **19** penetrating lid disk **18** in such a manner that an opening edge **20** of bung socket **19** protrudes from a top **21** of lid disk **18**.
- (9) As shown in FIG. 4 in particular, a bung plug **22** is screwed into bung socket **19**, bung plug **22** closing bung socket **19**. A sealing cap **23** is located on top of bung plug **22**, sealing cap **23** comprising a lid disk **24** and an edge skirt **25**, which is sleeve-shaped in the case at hand and extends essentially perpendicular to lid disk **11**, edge skirt **25** being connected to lid disk **24** via a predetermined breaking means **27**, which has predetermined breaking webs **26**, as is shown in FIG. 2 in particular.
- (10) As shown in FIG. 3 in particular, lid disk **24** is provided with a fastening means **29** on its underside **28**, fastening means **29** being a fastening ring in the case at hand and engaging a plug depression **30** of bung plug **19** in such a manner that a snap-lock means **38** formed on the lower end of fastening means **29** engages a mating snap-lock means **32** formed in plug depression **30**. As further shown in FIG. 3, lid disk **24** is provided with a support means, which is an edge web **33**, on its underside **28** radial inside adjacent to predetermined breaking means **27**, a lower edge **34** of edge web **33** resting on opening edge **20** of bung socket **19** disposed on top **21** of lid **17**.
- (11) As FIG. 4 in particular clearly shows, edge skirt **25** of the configuration example illustrated is connected to edge web **33** via predetermined breaking means **27**. Edge skirt **25** covers lower edge **34** of edge web **33**, which rests on opening edge **20**, since lower edge **34** is recessed relative to a lower edge **35** of edge skirt **25**.
- (12) If pressure is exerted on lid disk **24** as indicated by the force arrows in FIG. 4, a corresponding load is introduced into lid **17** through a force fit via edge web **33**, which rests on top **21** of lid disk **18** of lid **17**, more precisely on opening edge **20** of bung socket **19** in this case, without edge skirt **25** being loaded and reaction forces due to the loading of edge skirt **25** possibly acting on predetermined breaking means **27**. So the manipulation safeguard formed in this case by edge skirt **25**, which is connected to lid disk **18** via predetermined breaking means **27**, is not affected by external loads acting on lid disk **24** of sealing cap **23**.
- (13) On the other hand, sealing cap **23** is configured in such a manner that sealing cap **23** can be removed as a continuous unit from container closure **15**, in particular after predetermined breaking means **27** has been destroyed, without the support means, which is edge web **33** in this case,

preventing the removal. As illustrated in FIG. 5 in particular, a pull tab 36 formed on edge skirt 25 is connected to snap-lock means 38, which is formed on fastening means 29, via a connecting means 37, which has a lid strip 39, which extends from pull tab 36 via edge skirt 25 and edge web 33 into a central inner part 40 of lid disk 24 and is separated from adjacent portions of edge skirt 25, of edge web 33 and of inner part 40 of lid disk 24 by lateral predetermined breaking means 41 and 42. Lid strip 39 is connected to snap-lock means 38 of fastening means 29 via a pull web 43 in such a manner that when pull tab 36 is moved diagonally upward to the center of the lid disk, lid strip 39 is detached along predetermined breaking means 41 and 42, and snap-lock means 38, which is formed on fastening ring 29, is deformed radially inward via pull web 43, which is connected to inner part 40 of lid disk 24, with the result that sealing cap 23 can be removed from lid 17 as a continuous unit.

Claims

1. A container closure comprising: a container socket base wherein the container socket base is a lid for a container opening formed in a container wall, the lid comprising a bung socket which is disposed in a lid disk of the lid and which penetrates the lid disk of the lid; a sealing cap comprising a sealing cap lid disk which has a fastening means disposed on an underside of the sealing cap lid disk, the sealing cap lid disk provided with a snap-lock means for forming a snap-lock connection with a bung plug; and an edge skirt disposed at the circumference of the sealing cap lid disk and connected to the sealing cap lid disk via a predetermined breaking means, wherein the sealing cap lid disk has a support means for supporting the sealing cap lid disk on the container socket base, the support means being disposed radially inside adjacent to the predetermined breaking means on the underside of the sealing cap lid disk, and wherein the support means rests on a top of the lid disk of the lid and the edge skirt is spaced apart from the top of the lid disk of the lid.
 2. The container closure according to claim 1, wherein the bung socket has an opening edge which protrudes from a top of the lid in such a manner that the support means rests on the opening edge of the bung socket.
 3. The container closure according to claim 2, wherein the support means is an edge web formed on the underside of the sealing cap lid disk and disposed concentrically with the edge skirt.
 4. The container closure according to claim 3, wherein the edge web has a lower edge which is recessed relative to a lower edge of the edge skirt with respect to the underside of the sealing cap lid disk.
 5. The container closure according to claim 3, wherein the snap-lock means is connected to a pull tab disposed at the circumference of the sealing cap lid disk via a connecting means, the connecting means having a lid strip which is formed in the sealing cap lid disk and which is separated from adjacent areas of the edge skirt, the edge web, and an inner part of the sealing cap lid disk by a lateral predetermined breaking means and wherein the lid strip is connected to the snap-lock means via a pull web.
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